

elsearticle class for Typst

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MIT

Typst template for Elsevier article submissions

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`ELSEARTICLE` is a Typst template that aims to mimic the Elsevier article $\text{\LaTeX}$ class, a.k.a. `elsarticle.cls`, provided by Elsevier to format manuscript properly for submission to their journals.

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Part I

About

ELSEARTICLE is a Typst template that aims to mimic the Elsevier article L^AT_EX class, a.k.a. elsarticle.cls, provided by Elsevier to format manuscript properly for submission to their journals.

ELSEARTICLE is designed to be as close as possible to the original class, whose specification can be found in the [documentation](#)¹. The template is still in development and may not be fully compatible with all Elsevier journals.

This manual provides an overview of the features of the **ELSEARTICLE** template and how to use it.

To mimic the look and feel of the original elsarticle.cls, the following fonts must be used:

- **XITS**² and XITS Math for the best compatibility with the original elsarticle.cls;
- **STIX Two Text**³ and STIX Two Math as a reasonable alternative;
- New Computer Modern and New Computer Modern Math are also defined to avoid compilation errors if the previous fonts are not available.

The template is provided as is by the Typst community and is not affiliated with Elsevier.

¹<http://assets.ctfassets.net/o78em1y1w4i4/3ro3yQff1q67JHmLi1sAqV/1348e3852f277867230fc4b84a801734/elsdoc-1.pdf>

²<https://www.ctan.org/pkg/xits>

³<https://github.com/stipub/stixfonts>

Part II

Usage

II.1 Using elsearticle

To use the `ELSEARTICLE` template, you need to include the following line at the beginning of your `typ` file:

```
#import "@preview/elsearticle:2.1.0": *
```

II.2 Initializing the template

After importing `ELSEARTICLE`, you have to initialize the template by a `show` rule with the `#elsearticle` command. This function takes an optional argument to specify the title of the document.

```
#show: elsearticle.with(  
  ...  
)
```

`#elsearticle` takes the following arguments:

```
#elsearticle(  
  {title}: none,  
  {authors}: (),  
  {affiliations}: (),  
  {abstract}: none,  
  {journal}: none,  
  {keywords}: (),  
  {date}: none,  
  {format}: "preprint",  
  {paper}: "a4",  
  {numcol}: 1,  
  {line-numbering}: false  
)[body]
```

Argument

```
{title}: none
```

str

Title of the paper

Argument

`{authors}: ()`

array

Array containing the list of authors of the article. Each author is defined as a dictionary with the following keys:

- name: Name of the author (required) `str` | `content`
- affiliations: List of affiliations of the author (required) `array`
- corresponding: Corresponding author (optional, default: false) `bool`
- email: Email of the corresponding author (optional) `str` | `content`

```
authors: (
  (
    name: "J. Doe",
    affiliations: ("a", "b"),
    corresponding: true,
    email: "jdoe@univ.edu",
  ),
  (
    name: "J. Smith",
    affiliations: ("b",),
  ),
)
```

Argument

`{affiliations}: ()`

dictionary

Dictionary containing the list of affiliations of the article. Each affiliation is defined as a key-value pair, where the key is a `str` representing the affiliation ID and the value is a `str` | `content` giving the name and the address of the affiliation.

```
affiliations: (
  "a": [Institution A, City A, Country A],
  "b": [Institution B, City B, Country B],
)
```

If the paper has only one author, affiliations should be:

- In the `array` of `dictionary` :
 - `none`
 - an empty `dictionary` `()`
 - not defined at all

- (" ": [Institution name]) in the affiliations dictionary to conform with the Elsevier template requirements.

Argument

{abstract}: none

content

Abstract of the paper

Argument

{journal}: none

str

Name of the journal

Argument

{keywords}: ()

array

List of the keywords of the paper

Each element of the array is a str representing a keyword

keywords: ("Keyword 1", "Keyword 2")

Argument

{date}: none

datetime

Date to display. Defaults to today's date if omitted.

Argument

{format}: "review"

str

Format of the paper. Possible values are "preprint", "review", "1p", "3p" and "5p"

Argument

{paper}: "a4"

str

Paper size. Possible values are "a4", "ansi-a", "us-letter" and "us-legal"

If the value of paper is not in the list of possible values, it will be set to "a4" by default.

Argument

{numcol}: 1

number

Number of columns of the paper. Possible values are 1 and 2

According to the documentation of `elsearticle.cls` (see [here](#)⁴), the number of columns is related to the format of the paper:

- 1p: Single column only
- 3p: Single or double column possible
- 5p: Double column only

To avoid unexpected behaviors, the value of the `numcol` argument is set to 1 by default and restricted to 1 or 2.

⁴<https://assets.ctfassets.net/o78em1y1w4i4/3ro3yQff1q67JHmLi1sAqV/1348e3852f277867230fc4b84a801734/elsdoc-1.pdf>

Argument —

```
{line-numbering}: false bool
```

Enable line numbering in the document

II.3 Additional features

The `ELSEARTICLE` template provides additional features to help you format your document properly.

II.3.1 Appendix

The template allows you to create appendices using the `#appendix` environment. The appendices are then numbered with capital letters (A, B, C, etc.). Figures, tables and equations are numbered accordingly, e.g. Eq. (A.1).

To activate the appendix environment, all you have to do is to place the following command in your document:

```
#show: appendix

// Appendix content here
```

II.3.2 Subfigures

Subfigures are not built-in features of Typst, but the `ELSEARTICLE` template provides a way to handle them. It is based on the `SUBPAR` package that allows you to create subfigures and properly reference them.

To create a subfigure, you can use the following syntax:

```
#subfigure(
  figure(image("image1.png"), caption: []), <figa>,
  figure(image("image2.png"), caption: []), <figb>,
  columns: (1fr, 1fr),
  caption: [(a) Left image and (b) Right image],
  label: <fig>
)
```

The `#subfigure` function is a wrapper around the `#subpar.grid` function. The numbering is adapted to the context of the document (normal section or appendix).

II.3.3 Unnumbered sections and equations

In some cases, you may want to create unnumbered sections or equations. The `ELSEARTICLE` template provides a way to do this using the selector function and dedicated `label`. The latter can be used as follows:

```
// Unnumbered section
= Introduction <nonum-sec>

// Unnumbered equation
$

$$y = f(x)$$

$ <nonum-eq>
```

II.4 Cross-referencing

In `elsearticle`, the supplement of the `ref` function is set to “none” by default to avoid unexpected behavior when referencing more than one item. However, you can revert this setting by using the following syntax after the template definition:

```
#set ref(supplement: auto)
```


Part III

Dependencies

The `ELSEARTICLE` template depends on the following packages:

- `equate:0.3.2`: For equation handling and numbering.
- `subpar:0.2.2`: For subfigure handling and numbering.